

1.3 Application of Matrix Algebra: Input-Output Analysis

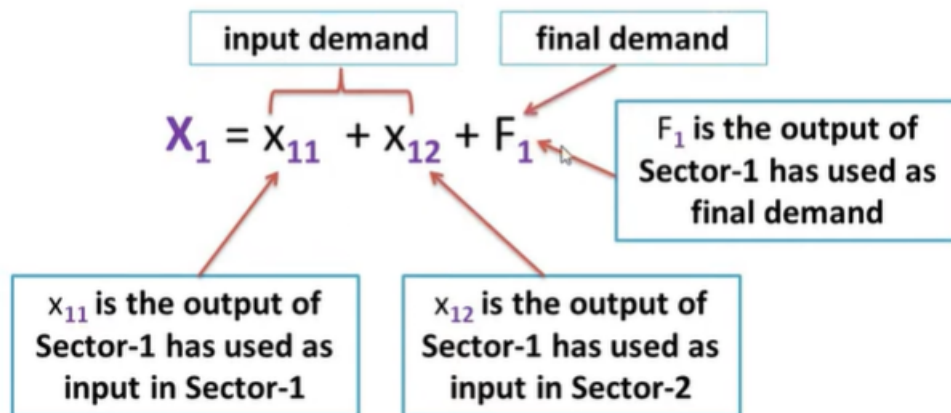
- Input-Output (I-O) Analysis is a quantitative data science method developed by Wassily Leontief that models interdependencies between economic sectors, tracking how outputs from one industry act as inputs for another. It uses matrix algebra to calculate direct and indirect impacts of demand changes across a system, making it essential for economic, environmental, and supply chain modelling.
- The Input-Output (I-O) Analysis earned Wassily Leontief the **Nobel Prize in Economics**. Thus, the input-output analysis is also known as the **Leontief input-output product**.
- The economy can be thought of as a composite entity of different sectors which are inter-dependent or inter-related. This mutual inter-dependence is empirically studied by using the tools of input-output analysis or input-output models.
- **Input:** Are raw materials or resources needed by producers or entrepreneurs for production.
- **Primary input** are the inputs from the producing sectors of the economy.
- The output of one sector will be consumed by the sectors which are producing (called intermediate demand) and by the sectors which are not producing (called Final demand/Consumption demand).

$$\text{Total output} = \text{Intermediate demand} + \text{Final demand}$$

Alternatively,

$$\text{Total output} = \text{input demand} + \text{Final demand}$$

For example in two sector economy,
say X_1 = output of Sector-1 and X_2 = output of Sector-2,
then



- **Goal:** The primary goal here is to set the amounts that each Sector must produce in order to satisfy both internal and external demand.

1.3.1 Assumptions of the input-output Models

1. The economy is divided into ‘n’ inter-dependent/inter-related sectors / industries.
2. Each sector/industry produces only **one** homogeneous commodity.
3. Each industry uses a fixed input ratio or factor combination for the production of its output.
4. **Proportionality Assumption.** The output of the i^{th} sector which is used as an input in the j^{th} sector will be proportional to the output of the j^{th} sector. That is

$$x_{ij} = a_{ij}x_j \quad (5)$$

For example, the output of the Agricultural sector, which is used as input for industrial sector, will be proportional to the output of the industrial sector. When the output of the industry increases, the demand for input will increase too.

Thus, a_{ij} indicates the amount of **output** of i^{th} sector used as **input** in the j^{th} sector to produce one or single unit of output of the j^{th} sector.

5. The proportionality coefficient a_{ij} is assumed to be **fixed** and is called **technological coefficient** or **Structural coefficient**.
6. The output of one sector is either used as an input of some other sector or it goes directly to **Final Consumption** or **Final demand**. In the Leontief Input-Output model, Final consumption represent external consumption (like households, government, exports) not used in production.
7. If an industry produces two different commodities or uses two different possible factor combinations, then the industry may be broken down into two separate industries.

1.3.2 Input-Output Flow Table

Consider the following input-output flow table for a two sector/industry analysis.

Sector(s)	input Agric	input industry	Final demand/Final Consumption	Total output
Agriculture	25 x_{11}	175 x_{12}	50 F_1	250 x_1
Industry	40 x_{21}	20 x_{22}	60 F_2	120 x_2
Labour services	10 x_{01}	40 x_{02}	0	50 x_0

The balancing equations are:

$$\begin{aligned}
x_{11} + x_{12} + F_1 &= x_1 \\
x_{21} + x_{22} + F_2 &= x_2 \\
x_{01} + x_{02} + 0 &= x_0
\end{aligned}$$

The three inputs: x_{11} , x_{21} and x_{01} produces the output x_1 (in Agriculture).
Therefore, $x_1 = f(x_{11}, x_{21}, x_{01})$.
Similarly, the inputs x_{12} , x_{22} and x_{02} produces the output x_2 (in Industry).
Therefore, $x_2 = f(x_{12}, x_{22}, x_{02})$.

Using the Proportionality Assumption, we can rewrite the balancing equation as follows.

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + F_1 &= x_1 \\
a_{21}x_1 + a_{22}x_2 + F_2 &= x_2
\end{aligned}$$

called the **structural equation** of a two-sector model (Agriculture and Industry, in this case).
From this equation, we can solve the input-output model using the Matrix Method.

1.3.3 Solving the input-Output Model System using Matrix Method

Consider the two-sector input-output model obtained earlier. Its structural equation is given by

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + F_1 &= x_1 \\
a_{21}x_1 + a_{22}x_2 + F_2 &= x_2
\end{aligned}$$

In Matrix form, we have

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \quad (6)$$

where,

$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is a **matrix of technological coefficients** or **consumption matrix**.

A matrix where each element a_{ij} represents the units of input required from sector i to produce one unit of output for sector j .

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is a vector of gross output
 $\begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$ is a vector of final/external demand

Equation (6) can further be transformed into

$$X = AX + F \quad (7)$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } F = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$
$$\begin{aligned} X - AX &= F \\ X[I - A] &= F \\ X &= [I - A]^{-1}F \end{aligned} \tag{8}$$

F = Final Demand
 I = Identity Matrix
 A = Matrix of Technological Coefficient
 $A]$ = Leontief's Matrix (should be non-singular)
 $^{-1}$ = Inverse of Leontief's Matrix

- $$\begin{aligned} X_1 &= x_{11} + x_{12} + x_{13} + \dots + x_{1n} + F_1 \\ X_2 &= x_{21} + x_{22} + x_{23} + \dots + x_{2n} + F_2 \\ X_3 &= x_{31} + x_{32} + x_{33} + \dots + x_{3n} + F_3 \\ &\vdots \\ X_n &= x_{n1} + x_{n2} + x_{n3} + \dots + x_{nn} + F_n \end{aligned}$$

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$$\begin{aligned}
X_1 &= a_{11} X_1 + a_{12} X_2 + a_{13} X_3 + \dots + a_{1n} X_n + F_1 \\
X_2 &= a_{21} X_1 + a_{22} X_2 + a_{23} X_3 + \dots + a_{2n} X_n + F_2 \\
X_3 &= a_{31} X_1 + a_{32} X_2 + a_{33} X_3 + \dots + a_{3n} X_n + F_3 \\
&\dots \quad \dots \quad \dots \quad \dots \\
&\dots \quad \dots \quad \dots \quad \dots \\
X_n &= a_{n1} X_1 + a_{n2} X_2 + a_{n3} X_3 + \dots + a_{nn} X_n + F_n
\end{aligned}$$

- In matrix form

$$\begin{array}{c}
\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \times \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ \vdots \\ X_n \end{bmatrix} + \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ \vdots \\ \vdots \\ F_n \end{bmatrix} \\
\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
X \quad = \quad A \quad \quad \quad X \quad + \quad F
\end{array}$$

$$X = AX + F$$

$$X - AX = F$$

$$(I - A)X = F \quad \text{or} \quad X = (I - A)^{-1}F$$

Example 1.3.1. Consider the following input-output table for a two-sector economy.

Sector(s)	input for Agric	input for industry	Final demand	Total output
Agriculture	300 x_{11}	600 x_{12}	100 F_1	1000 x_1
Industry	400 x_{21}	1200 x_{22}	400 F_2	2000 x_2

Suppose we are to calculate the new gross output for the model so as to meet a new final demand F , given by

$$F = \begin{bmatrix} 200 \\ 800 \end{bmatrix}$$

we proceed as follows:

[step 1:] Calculate the matrix of technical coefficients A . Using the proportionality assumptions,

$$\begin{aligned} a_{11} &= \frac{x_{11}}{x_1} = \frac{300}{1000} = 0.3; & a_{12} &= \frac{x_{12}}{x_2} = \frac{600}{2000} = 0.3; \\ a_{21} &= \frac{x_{21}}{x_1} = \frac{400}{1000} = 0.4; & a_{22} &= \frac{x_{22}}{x_2} = \frac{1200}{2000} = 0.6; \end{aligned}$$

Thus, the matrix of technical coefficients, A , is

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 0.3 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

[step 2:] Use the formula $X = [I - A]^{-1}F$ to obtain the required gross output. In this case, $F = \begin{bmatrix} 200 \\ 800 \end{bmatrix}$.

Also,

$$[I - A] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.3 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} 0.7 & -0.3 \\ -0.4 & 0.4 \end{bmatrix}$$

$$[I - A]^{-1} = \frac{Adj[I - A]}{|I - A|} \quad \text{where,}$$

$$|I - A| = (0.7 \times 0.4) - (-0.4 \times -0.3) = 0.16$$

Matrix of minors:

$$\begin{aligned} \text{minor of } 0.7 &= 0.4; & \text{minor of } -0.3 &= -0.4; \\ \text{minor of } -0.4 &= -0.3; & \text{minor of } 0.4 &= 0.7 \end{aligned}$$

$$\text{Matrix of minors} = M_{ij} = \begin{bmatrix} 0.4 & -0.4 \\ -0.3 & 0.7 \end{bmatrix}$$

Matrix of cofactor: $C_{ij} = (-1)^{i+j}M_{ij}$

$$\begin{array}{ll} \text{cofactor of } 0.7 = (-1)^{1+1}(0.4) = 0.4; & \text{cofactor of } -0.3 = (-1)^{1+2}(-0.4) = 0.4; \\ \text{cofactor of } -0.4 = (-1)^{2+1}(-0.3) = 0.3; & \text{cofactor of } 0.4 = (-1)^{2+2}(0.7) = 0.7 \end{array}$$

$$\text{Matrix of cofactors} = C_{ij} = \begin{bmatrix} 0.4 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$\text{Adj}(A) = C_{ij}^T = \begin{bmatrix} 0.4 & 0.3 \\ 0.4 & 0.7 \end{bmatrix}$$

Therefore,

$$X = [I - A]^{-1}F = \frac{1}{0.16} \begin{bmatrix} 0.4 & 0.3 \\ 0.4 & 0.7 \end{bmatrix} \begin{bmatrix} 200 \\ 800 \end{bmatrix} = \begin{bmatrix} 2000 \\ 4000 \end{bmatrix}$$

That is, 2000 units is the gross output of the Agriculture sector and 4000 units is the gross output of the industrial sector, needed to meet the final demand, $F = \begin{bmatrix} 200 \\ 800 \end{bmatrix}$.

Example 1.3.2. Suppose that there are only three industries in the economy and one primary input and the input co-efficient matrix follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0.2 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.2 \\ 0.1 & 0.3 & 0.2 \end{bmatrix}$$

If the specific final-demand vector, say the final output target of a development program happens to

be $F = \begin{bmatrix} 10 \\ 5 \\ 6 \end{bmatrix}$, what are the solution output levels for the three industries.

Solution:

Given: Input coefficient matrix, A (Matrix of Technological Coefficients) and final-demand, F .

To find: Output levels for the three industries.

Using the Input-Output model equation,

$$X = [I - A]^{-1}F$$

[Step 1:] Computing the Leontief matrix, $[I - A]$,

$$[I - A] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.2 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.2 \\ 0.1 & 0.3 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.3 & -0.2 \\ -0.4 & 0.9 & -0.2 \\ -0.1 & -0.3 & 0.8 \end{bmatrix}$$

[Step 2:] Computing the inverse of $[I - A]$, i.e. $[I - A]^{-1}$

$$[I - A]^{-1} = \frac{1}{|I - A|} \cdot Adj(I - A)$$

$$\begin{aligned} |I - A| &= 0.8 \begin{vmatrix} 0.9 & -0.2 \\ -0.3 & 0.8 \end{vmatrix} - (-0.3) \begin{vmatrix} -0.4 & -0.2 \\ -0.1 & 0.8 \end{vmatrix} + (-0.2) \begin{vmatrix} -0.4 & 0.9 \\ -0.1 & -0.3 \end{vmatrix} \\ &= 0.8(0.66) + 0.3(-0.34) - 0.2(0.12 + 0.09) \\ &= \mathbf{0.384} \end{aligned}$$

$$\text{Co-factor of } a_{11}, \text{ i.e. } 0.8 = + \begin{vmatrix} 0.9 & -0.2 \\ -0.3 & 0.8 \end{vmatrix} = \mathbf{0.66};$$

$$\text{Co-factor of } a_{12}, \text{ i.e. } -0.3 = - \begin{vmatrix} -0.4 & -0.2 \\ -0.1 & 0.8 \end{vmatrix} = \mathbf{0.34}$$

$$\text{Co-factor of } a_{13}, \text{ i.e. } -0.2 = + \begin{vmatrix} -0.4 & 0.9 \\ -0.1 & -0.3 \end{vmatrix} = \mathbf{0.21}$$

$$\text{Co-factor of } a_{21}, \text{ i.e. } -0.4 = - \begin{vmatrix} -0.3 & -0.2 \\ -0.3 & 0.8 \end{vmatrix} = \mathbf{0.30};$$

$$\text{Co-factor of } a_{22}, \text{ i.e. } 0.9 = + \begin{vmatrix} 0.8 & -0.2 \\ -0.1 & 0.8 \end{vmatrix} = \mathbf{0.62}$$

$$\text{Co-factor of } a_{23}, \text{ i.e. } -0.2 = - \begin{vmatrix} 0.8 & -0.3 \\ -0.1 & -0.3 \end{vmatrix} = \mathbf{0.27}$$

$$\text{Co-factor of } a_{31}, \text{ i.e. } -0.1 = + \begin{vmatrix} -0.3 & -0.2 \\ 0.9 & -0.2 \end{vmatrix} = \mathbf{0.24};$$

$$\text{Co-factor of } a_{32}, \text{ i.e. } -0.3 = - \begin{vmatrix} 0.8 & -0.2 \\ -0.4 & -0.2 \end{vmatrix} = \mathbf{0.24}$$

$$\text{Co-factor of } a_{33}, \text{ i.e. } 0.8 = + \begin{vmatrix} 0.8 & -0.3 \\ -0.4 & 0.9 \end{vmatrix} = \mathbf{0.60}$$

$$\text{Co-factor Matrix of } [I - A] = C_{ij} = \begin{bmatrix} 0.66 & 0.34 & 0.21 \\ 0.30 & 0.62 & 0.27 \\ 0.24 & 0.24 & 0.60 \end{bmatrix}$$

$$\text{Adjoint Matrix, } Adj(I - A) = C_{ij}^T = \begin{bmatrix} 0.66 & 0.30 & 0.24 \\ 0.34 & 0.62 & 0.24 \\ 0.21 & 0.27 & 0.60 \end{bmatrix}$$

$$\text{Now } [I - A]^{-1} = \frac{1}{|I - A|} \cdot Adj(I - A)$$

$$[I - A]^{-1} = \frac{1}{0.384} \begin{bmatrix} 0.66 & 0.30 & 0.24 \\ 0.34 & 0.62 & 0.24 \\ 0.21 & 0.27 & 0.60 \end{bmatrix}$$

[Step 3:] Next, compute the output X , where $X = (I - A)^{-1}F$

$$X = \frac{1}{0.384} \begin{bmatrix} 0.66 & 0.30 & 0.24 \\ 0.34 & 0.62 & 0.24 \\ 0.21 & 0.27 & 0.60 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 6 \end{bmatrix} = \frac{1}{0.384} \begin{bmatrix} 9.54 \\ 7.94 \\ 7.05 \end{bmatrix} = \begin{bmatrix} 24.84 \\ 20.68 \\ 18.36 \end{bmatrix}$$

Therefore, the output levels for the first industry, 2nd industry and 3rd industry are 24.84 units, 20.68 units and 18.36 units, respectively.

Exercise 2.

1. A town has a merchant, a baker and a farmer. To produce \$1 of output, the merchant requires \$0.30 of baked goods and \$0.40 worth of the farmer's products. To produce \$1 of output, the baker requires \$0.50 worth of goods from the merchant, \$0.10 worth of its own goods and \$0.30 worth of the farmer's products. To produce \$1 of output, the farmer requires \$0.30 worth of goods from the merchant, \$0.20 worth of goods from the baker and \$0.30 worth of its own goods. How much should each produce to meet consumer demand of \$20,000 worth of output from the merchant, \$15,000 worth of goods from the baker and \$18,000 worth of goods from the farmer? **Ans:**

$$A = \begin{bmatrix} 0 & 0.5 & 0.3 \\ 0.3 & 0.1 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{bmatrix} \quad \text{and} \quad F = \begin{bmatrix} 20 \\ 15 \\ 18 \end{bmatrix}$$

\$85034.50 from the merchant; **\$68,000** from the bakers and **\$103,448.30** from the farmer.

2. Consider an open Leontief model with input matrix T given by

$$T = \begin{bmatrix} 0.2 & 0.3 & 0.2 \\ 0.4 & 0.1 & 0.2 \\ 0.1 & 0.3 & 0.2 \end{bmatrix}$$

Show that if $A = [I - T]$ then A^{-1} is approximated by

$$A^{-1} = \frac{1}{0.38} \begin{bmatrix} 0.66 & 0.30 & 0.24 \\ 0.34 & 0.64 & 0.24 \\ 0.21 & 0.27 & 0.60 \end{bmatrix}$$

where the entries are rounded to two decimal places.

3. Show that the model with input matrix T described in 2 is feasible and compute the output x_1 , x_2 and x_3 associated with the open demands

$$F_1 = 10, F_2 = 5, F_3 = 6$$

4. Example

Example:

$$A = \begin{bmatrix} 0.3 & 0.2 & 0.4 \\ 0 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.2 \end{bmatrix}; \quad F = \begin{bmatrix} 300 \\ 500 \\ 400 \end{bmatrix}$$

$$\therefore I - A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.3 & 0.2 & 0.4 \\ 0 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{bmatrix}$$

The Leontief input-output model is $(I - A) X = F$

$$\begin{bmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 300 \\ 500 \\ 400 \end{bmatrix}$$

Using Cramer's rule,

$$\begin{bmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 300 \\ 500 \\ 400 \end{bmatrix}$$

$$X_1 = \frac{\begin{vmatrix} 300 & -0.2 & -0.4 \\ 500 & 0.8 & -0.1 \\ 400 & -0.2 & 0.8 \end{vmatrix}}{\begin{vmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{vmatrix}} = \frac{442}{0.4} = 1105$$

$$X_2 = \frac{\begin{vmatrix} 0.7 & 300 & -0.4 \\ 0 & 500 & -0.1 \\ -0.1 & 400 & 0.8 \end{vmatrix}}{\begin{vmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{vmatrix}} = \frac{290}{0.4} = 725$$

$$X_3 = \frac{\begin{vmatrix} 0.7 & -0.2 & 300 \\ 0 & 0.8 & 500 \\ -0.1 & -0.2 & 400 \end{vmatrix}}{\begin{vmatrix} 0.7 & -0.2 & -0.4 \\ 0 & 0.8 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{vmatrix}} = \frac{328}{0.4} = 820$$

Thus,

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 1105 \\ 725 \\ 820 \end{bmatrix}$$

1.3.4 Interpretation of entries of $(I - A)^{-1}$

- Observe that the equation

$$X = (I - A)^{-1}F$$

gives the relationship between the external demand, F , and the amount that the Sectors must produce, X .

- However the entries in $(I - A)^{-1}$ itself have their own interpretation. To understand what they mean, suppose that we have some A and some X . For the sake of simplicity assume A is 3×3 .
- Let's investigate what happens if the external demand for Product 1 changes by +1. The new external demand is

$$F + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

How does X change? Well

$$\begin{aligned}
 X_{new} &= (I - A)^{-1} \left(F + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) \\
 &= (I - A)^{-1} F + (I - A)^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\
 X_{new} &= X + (I - A)^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}
 \end{aligned}$$

- Note that the vector

$$(I - A)^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

contains the

first column of $(I - A)^{-1}$ which shows us that the first column in $(I - A)^{-1}$ indicates how the production must change in each of the three Sectors if the external demand in the first Sector changes by +1.

- Consider this example:

Example 1.3.3. Suppose

$$A = \begin{bmatrix} 0.1 & 0.15 & 0.12 \\ 0.2 & 0 & 0.3 \\ 0.25 & 0.4 & 0.2 \end{bmatrix}$$

If we calculate, we find:

$$(I - A)^{-1} = \begin{bmatrix} 1.2660 & 0.3128 & 0.3072 \\ 0.4375 & 1.2850 & 0.5473 \\ 0.6144 & 0.7400 & 1.6200 \end{bmatrix}$$

So now if the external demand for Product 1 changes by +1 the production in the three Sectors must change by +1.266, +0.4375 and +0.6144, respectively.

- **Fact:** If the external demand for Sector j changes by +1 then column j of $(I - A)^{-1}$ indicates how production must change in all Sectors in order to compensate.
- **Fact:** If the external demand for Sector j changes by +1 then the i^{th} entry of column j of $(I - A)^{-1}$ indicates how the production in Sector i must change in order to compensate.

- **Fact:** This change is linear in nature in that if the external demand for Sector j changes by, for example, +2 then we can simply double column j .

Example 1.3.4. *If an economy with five Sectors has consumption matrix*

$$A = \begin{bmatrix} 0.0200 & 0 & 0.1000 & 0.0200 & 0 \\ 0.06 & 0 & 0.0400 & 0 & 0.0700 \\ 0 & 0.0150 & 0 & 0.0110 & 0 \\ 0 & 0.0220 & 0 & 0 & 0.0800 \\ 0.0300 & 0 & 0.0320 & 0.0100 & 0 \end{bmatrix}$$

Then

$$(I - A)^{-1} = \begin{bmatrix} 1.0206 & 0.0020 & 0.1022 & 0.0216 & 0.0019 \\ 0.0634 & 1.0008 & 0.0486 & 0.0025 & 0.0703 \\ 0.0010 & 1.0153 & 1.0008 & 0.0110 & 0.0020 \\ 0.0039 & 0.0221 & 0.0039 & 1.0009 & 0.0816 \\ 0.0307 & 0.0008 & 0.0351 & 0.0110 & 1.0009 \end{bmatrix}$$

so for example if the external demand for the fourth Sector changes by +1 then the production of the five Sectors will need to change by

$$\text{Column 4} = \begin{bmatrix} 0.0216 \\ 0.0025 \\ 0.0110 \\ 1.0009 \\ 0.0110 \end{bmatrix}$$

respectively. For example Sector 1 must change production by +0.02155, Sector 2 must change production by +0.002506, and so on.

- Notice that the production of Sector 4 must go up by more than 1. This makes sense since it must produce enough to cover the new external demand for Product 4 plus enough to cover the internal demands of all the Sectors (including itself) as they work together in harmony to increase production.
- On the other hand, if the external demand for the Sector 4 product changes by -2 then the production of the five Sectors will need to change by

$$-2(\text{Column 4}) = -2 \begin{bmatrix} 0.0216 \\ 0.0025 \\ 0.0110 \\ 1.0009 \\ 0.0110 \end{bmatrix} = \begin{bmatrix} -0.0431 \\ -0.0050 \\ -0.0221 \\ -2.0019 \\ -0.0220 \end{bmatrix}$$

respectively.